

What I Built

In this project, I developed two main tasks: **Image Processing** and **Webcam Color Detection**.

Task 1: Image Processing I created a set of functions that allow for loading, processing, and saving images. The image transformations include:

- **Grayscale Conversion:** Turning images into black and white.
- **Gaussian Blur:** Smoothing the image to reduce noise.
- **Canny Edge Detection:** Detecting and highlighting edges in the image.

I also added functionality to process all images in a folder, applying these transformations to each one automatically.

Task 2: Webcam Color Detection This system detects the closest color from a list of predefined colors when you click on the webcam feed. It involves:

- Loading color names and RGB values from a CSV file.
- Comparing the clicked color with the loaded color list to find the closest match using mathematical calculations.
- Displaying the closest color's name and RGB values on the webcam feed.

Challenges I Faced

1. Image Processing:
 - I faced a problem ensuring images are loaded and saved correctly, especially when handling paths and directories.
 - I faced a problem to understand the functionality of the function Like Canny.
2. Webcam Color Detection:
 - I faced a problem to enable a webcam (inbuilt)
 - I faced a problem to understand the functionality of different function of the openCV.
 - I faced a problem of managing the real-time webcam feed and ensuring smooth operation without delays.
 - I faced a problem of finding closer color as Calculating the Euclidean distance between the clicked color and the colors in the CSV file required precise math for accurate results.

What I Would Improve if I Had More Time

1. Image Processing:
 - **Error Handling:** Improve how errors are handled, especially when images are missing or not in the correct format.
 - **Customization:** Allow users to modify settings, such as blur size or edge detection thresholds, to suit different needs.
2. Webcam Color Detection:
 - **Multiple Color Detection:** Enable detection of multiple colors at once, allowing for more complex color identification.
 - **Performance Optimization:** Optimize the webcam feed to reduce lag and ensure smoother operation, even with larger color lists.
 - **GUI Interface:** Build a simple graphical user interface (GUI) for users to interact with the system more easily, allowing them to load colors and use webcam features.

If I had more time then I would learn about the computer vision and openCV library and also I explore the different functions of the openCV and much more.